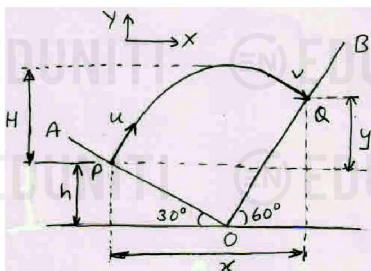


SUBJECTIVE QUESTIONS

1. A point moves rectilinearly with a resisting acceleration that depends on the velocity v of the particle as $a = -\alpha v^{1/2}$ where α is a positive constant. At $t = 0$, the velocity is v . What distance will it traverse before it stops? What time will it take to cover that distance?

2. A stone is projected from the point on a ground in such a direction so as to hit a bird on the top of a telegraph post of height h and then attain the maximum height $2h$ above the ground. If at the instant of projection, the bird were to fly away horizontally with a uniform speed, find the ratio between the horizontal velocities of the bird and the stone, if the stone still hits the bird while descending.

3. Two inclined planes OA and OB having inclination (with horizontal) 30° and 60° respectively, intersect each other at O as shown in fig.



A particle is projected from point P with velocity $u = 10\sqrt{3} \text{ ms}^{-1}$ along a direction perpendicular to plane OA. If the particle strikes plane OB perpendicularly at Q, calculate the

- time of flight.
- speed with which particle strikes the plane OB.
- vertical height h of P from O.
- distance PQ.
- maximum height from O, attained by the particle.

4. A particle is projected with a velocity of 30 m/s at an angle 60° above the horizontal on a slope of inclination 30° . Find its range, time of flight and angle of hit. (Take $g = 10 \text{ m/s}^2$)

5. Two guns, situated on the top of a tower, fire one shot each with the same speed $5\sqrt{3} \text{ m/s}$ at some interval of time. The first gun fires upward at an angle of 60° with the horizontal and the second gun fires horizontally. The shots collide in air at point P. Find

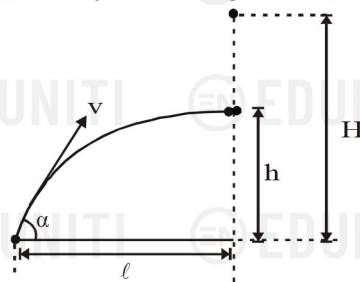
- the coordinates of point P.
- the time interval between the firings.

Take the origin of the coordinate system at the point of the firing and trajectories in the x-y plane.

6. The speed of the particle when it is at greatest height is A body is $\sqrt{2/5}$ times the speed when it is at half of its greatest height. Find the angle of projection.

7. A body is thrown at an angle θ with the horizontal such that it attains a speed equal to $\sqrt{2/3}$ times the speed of projection when the body is at half to its maximum height. Find the angle θ

8. A body falls freely from some altitude H . At the moment the first body starts falling, another body is thrown from the ground which collides with the first at an altitude $h = H/2$. The horizontal distance covered by the second body is l . Find the initial velocity and the angle at which it was thrown.



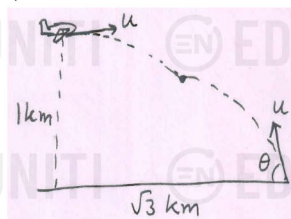
9. At what angle α above the horizon should a stone be thrown from a steep bank for it to fall into the water as far as possible from the bank? The height of the bank $h = 20 \text{ m}$ and the initial velocity of the stone $u = 15 \text{ ms}^{-1}$. (Take $g = 10 \text{ ms}^{-2}$)

10. Two swimmers start a race. One who reaches point C first on the other bank wins the race. A makes his strokes in a direction of 37° to the river flow with velocity 5 km/hr relative to water. B makes his strokes in a direction 127° to the river flow with same relative velocity. River is flowing with speed of 2 km/hr and is 100 m wide. Who will win the race? Compute the time taken by A and B to reach the point C if the speeds of A and B on the ground are 8 km/hr and 6 km/hr respectively.

11. During a rainy day, rain is falling vertically with a velocity 2 m/s . A boy at rest starts his motion with a constant acceleration of 2 m/s^2 along a straight road. Find the rate at which the angle of the axis of umbrella with vertical should be changed so that the rain always falls parallel to the axis of the umbrella.

12. An aircraft is flying horizontally with a constant velocity $u = 200 \text{ m/s}$, at a height $= 1 \text{ km}$ above the ground. At the moment shown, a bomb is released from the aircraft and the cannon-gun below fires a shell with initial speed of 200 m/s , at some angle θ .

- For what value of θ will the projectile shell destroy the bomb in mid-air?
- If the value of θ is 53° , find the minimum distance between the bomb and the shell as they fly past each other. (Take $\sin 53^\circ = 4/5$.)



13. A balloon moves up vertically such that if a stone is thrown from it with a horizontal velocity v_0 relative to it, the stone always hits the ground at a fixed point $2v_0^2/g$ horizontally away from it. Find the height of the balloon as a function of time.

PARAGRAPH TYPE**Paragraph for Question 14 to 16**

From a tower of height 40 m, two bodies are simultaneously projected horizontally in opposite directions, with velocities 2 m s^{-1} and 8 m s^{-1} , respectively. ($g = 10 \text{ m s}^{-2}$)

14. The time taken for the velocity vectors of two bodies to become perpendicular to each other, is
(a) 0.1 s (b) 0.2 s
(c) 0.4 s (d) 0.8 s
15. The horizontal distance between two bodies, when their velocity are perpendicular to each other, is
(a) 1 m (b) 0.5 m
(c) 2 m (d) 4 m
16. The time taken for the displacement vectors of two bodies to become perpendicular to each other is
(a) 0.1 s (b) 0.2 s
(c) 0.8 s (d) 0.6 s

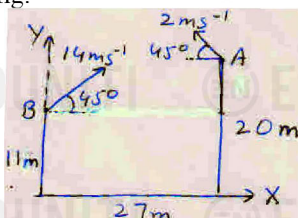
Paragraph for Question 17 to 18

A rabbit is hidden in grass at point A near a pond. A dog is drinking water in the pond at point B. The coordinates of points A and B are (0, 0) and (46 m, 28 m), respectively. At the same instant, rabbit starts to run with a constant velocity $(7.5\hat{i} + 10\hat{j}) \text{ m s}^{-1}$ and dog starts to run with constant speed in a straight line and catches rabbit in time t .

17. If x and y -components of velocities of dog for catching the rabbit are v_1 and v_2 , respectively, which of following relations is correct?
(a) $7.5t - 46 = v_1 t$ (b) $10t - 28 = v_2 t$
(c) Both (a) and (b) (d) Neither (a) nor (b)
18. If it runs with minimum speed, the minimum time in which the dog could catch the rabbit is
(a) 4 s (b) 5.25 s
(c) 5 s (d) 4.6 s

Paragraph for Question 19 to 21

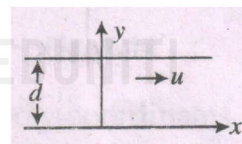
Two particles are thrown simultaneously from points A and B with velocities $u_1 = 2 \text{ m s}^{-1}$ and $u_2 = 14 \text{ m s}^{-1}$, respectively, as shown in fig.



19. The relative velocity of B as seen from A is
(a) $8\sqrt{2}\hat{i} + 6\sqrt{2}\hat{j}$ (b) $4\sqrt{2}\hat{i} + 3\sqrt{3}\hat{j}$
(c) $3\sqrt{5}\hat{i} + 2\sqrt{3}\hat{j}$ (d) $3\sqrt{2}\hat{i} + 4\sqrt{3}\hat{j}$
20. Minimum separation between A and B is
(a) 3 m (b) 6 m
(c) 9 m (d) 12 m
21. Time after which the separation between A and B is minimum is
(a) 2.7 s (b) $1.35\sqrt{2} \text{ s}$
(c) 2 s (d) 3 s

Paragraph for Question 22 to 24

A river of width d is flowing with velocity u . A boat starts from rest and with respect to water, moves perpendicular to the river with an acceleration of $a = 5t$, where t is time. The boat starts from point (1, 0) of the coordinate system as shown in fig.



22. Find the equation of trajectory of the boat.
(a) $y = \frac{6}{5} \left(\frac{x}{u} \right)^3$ (b) $y = \frac{5}{6} \left(\frac{x}{u} \right)^3$
(c) $y = \frac{5}{6} = u \left(\frac{x-1}{u} \right)^3$ (d) $y = \frac{6}{5} \left(\frac{x-1}{u} \right)^3$
23. Find the drift of the boat when it is in the middle of the river.
(a) $u \left(\frac{3d}{5} \right)^{1/3}$ (b) $u \left(\frac{3d}{5} \right)^{1/3} + 1$
(c) $u \left(\frac{5d}{6} \right)^{1/3}$ (d) $u \left(\frac{5d}{6} \right)^{1/3} + 1$
24. Obtain the total time taken to cross the river.
(a) $(3d/5)^{1/3}$ (b) $(6d/5)^{1/3}$
(c) $(5d/3)^{1/3}$ (d) $(5d/6)^{1/3}$

Paragraph for Question 25 to 27

A projectile is projected with some initial velocity and some initial angle of projection. Wind is also blowing, due to which constant horizontal retardation a is imparted to the particle in the plane of motion. It is found that the particle is at same height at two different time t_1 and t_2 and particle is at same horizontal distance at two different time t_3 and t_4 .

25. Angle of projection of particle is

(a) $\tan^{-1} \left(\frac{t_1 + t_2}{t_3 + t_4} \right)$ (b) $\tan^{-1} \left[\frac{a(t_1 + t_2)}{g(t_3 + t_4)} \right]$



(c) $\tan^{-1} \left[\frac{g(t_1 + t_2)}{a(t_3 + t_4)} \right]$ (d) None of these

26. Maximum height of the particle is

(a) $\frac{g}{8}(t_1 + t_2)^2$ (b) $\frac{g}{4}(t_1 + t_2)^2$
(c) $\frac{g}{2}(t_1 + t_2)^2$ (d) None of these

27. Range of the projectile is

(a) $\frac{1}{2}a(t_1 + t_2)^2$ (b) $\frac{1}{2}a(t_1 + t_2)(t_3 + t_4 - t_1 - t_2)$
(c) $\frac{1}{2}a(t_3 + t_4 - t_1 - t_2)^2$
(d) can't be determined